

One Tick, One Link:

A Non-Autoregressive Challenge–Response Memory in Which Depth Is Bought with the World’s Silence

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Abstract

We report a sequence learner that is not autoregressive. It is *challenge–response*: one input per tick, one output per tick, and intervals in which the input is the identity and the model keeps emitting. Its two operations are inverses selected by the world — speech *binds* an observation into a superposed store, silence *unbinds* one link out of it — so that depth is neither a layer count nor a hyperparameter but the number of quiet ticks the world grants. Several partial responses mature in parallel and the one that has matured is said. There is no reward, no label, and no gradient across a memory write; the only external quantity is prequential codelength. We verify in isolation that a chain is traversed one link per quiet tick and halts at its end, derive and confirm the store’s operating region $d > 2k \ln V$, show that binding is invertible only for unitary codebooks, and show that locality-sensitive addressing makes a near miss undetectable by construction. On a synthetic continual stream retrieval occurs and holds, and parallel responses are load-bearing. We rule on none of the components that measure net-negative, because the explanation for all of them rests on a property of our source rather than of the world, and because a replayed file cannot test a system for which time is a resource.

1 Introduction

1.1 Three commitments of the autoregressive frame

The dominant frame for sequence learning factorises $p(x_{1:T}) = \prod_t p(x_t | x_{<t})$, feeds a prefix, and predicts the next symbol [Elman, 1990, Vaswani et al., 2017]. It has been extraordinarily productive, and it carries three commitments that are easy to stop seeing because everything around them assumes the same ones.

The objective is next-symbol prediction. Every metric derived from it inherits that shape. A system evaluated by per-symbol codelength is being asked how well it does the thing autoregression is for, whatever else it may have been built to do.

Computation per symbol is fixed. A forward pass costs what it costs whether the world gives the model a microsecond or an hour. Depth is a property of the architecture, chosen at design time. Recent stateful models [Gu and Dao, 2023] keep a running state rather than a window, but the computation per token is still fixed and the objective is still next-symbol.

The context is a prefix. An exact, recent, ordered window. Long-range association is bought by widening it [Beltagy et al., 2020, Dai et al., 2019] or by retrieving into it [Borgeaud et al.,

2022, Weston et al., 2015]. Association at distance is thereby made a question about how much text you can hold, rather than about what you have stored.

For a system meant to run in a real temporal world — embodied [Brooks, 1991], or agentic over long horizons — all three sit badly. There is no prefix boundary in a room. Computation that cannot use extra time cannot exploit a lull, and lulls are most of what a real stream contains. And an exact window is the wrong idealisation for a background that in any organism is a fuzzy, multi-timescale trace rather than a transcript [Brainerd and Reyna, 2002].

1.2 What we put in their place

We take challenge–response as the primitive. There is no “a challenge arrives, now respond” — that phrasing is already the autoregressive shape, one prompt one completion. There is a stream, and every tick carries one input and produces one output. Some inputs are the world’s tokens; in the intervals between them the input is the identity and the model still emits, iterating on memory. The model’s own emissions are part of its context, interleaved with the world’s: distinguished from them, and combined with them.

The picture we have kept in view is two people talking. While the other is still speaking you have already begun to think; when they stop, or just as they stop, you begin to say something. That interval is the gap. What is forming in it is *not one trajectory* — several half-formed replies mature together and one of them is what comes out. Each clause of that sentence turned out to be load-bearing, and each had to be recovered separately after a run in which the mechanism measured inert (§8).

The mechanism follows from taking the identity input seriously. If speech means “there is something to combine”, then silence means “there is not”, and what is called for is the inverse operation. Binding writes; unbinding reads; the tick already says which. Nothing is scheduled, and — crucially — nothing needs to know whether a question is outstanding.

1.3 What is borrowed

Storage is content-addressed through a fixed map that no later learning can move, following the position argued in Yang [2026]: allocation buys freedom from interference and pays for it in retrieval difficulty, and the weakness of the family is that discrimination between addresses decays as addresses crowd. That trade organises our design, and one of our results is that in our implementation half of the crowding penalty was self-inflicted (§4.3).

The representation is holographic [Plate, 1995] in the vector symbolic tradition [Smolensky, 1990, Kanerva, 1988, 2009, Gayler, 2003, Eliasmith, 2013, Kleyko et al., 2023]: binding by circular convolution, reading by circular correlation, cleanup against a codebook. The pairing of an associative store with small-world routing follows Watts and Strogatz [1998]. The readout is a delta rule [Widrow and Hoff, 1960, Rosenblatt, 1958]. The continual setting — regimes arriving, running, departing permanently, with no task identifier and no boundary signal — is the one in which catastrophic interference was characterised [McCloskey and Cohen, 1989, French, 1999] and to which complementary-systems and regularisation accounts respond [McClelland et al., 1995, Kirkpatrick et al., 2017, Parisi et al., 2019].

1.4 Contributions

1. A tick in which binding and unbinding are two halves of one loop selected by whether the world spoke, so that gap periods, self-output-as-context, and depth-from-time are consequences of a single rule rather than three mechanisms (§3, Figure 1).
2. An isolated demonstration that a chain is traversed one link per quiet tick, halts at its end, and is unharmed by surplus silence (§4, Figure 2).

3. The operating region of a superposed store, $d > 2k \ln V$, derived and confirmed, with the observation that circular correlation inverts convolution only for unitary codebooks — a Gaussian codebook loses almost half the signal with a single item stored (§4.2).
4. A structural result: locality-sensitive addressing makes a near miss undetectable *by construction*, because a bank’s occupants have keys close to the query. Addressing needs same-content-same-bank and gains nothing from similar-content-similar-bank (§4.3).
5. An explicit account of what a constructed source cannot settle, and of why a replayed stream cannot test a model for which time is a resource (§7, §9).

2 Philosophical commitments

These are not decoration. Each translates into a constraint that the implementation is checked against, and several of our negative results are consequences of holding to them.

2.1 Intelligence as the organisation of memory, not as optimisation against a scalar

The founding position of this work is that intelligence is an emergence of memory rather than a product of reward. The contrast is with the framing in which behaviour is whatever maximises a scalar [Sutton and Barto, 2018, Silver et al., 2021]; here there is no reward, no value function, and no policy. It is closer to the tradition in which memory is the computational substrate rather than a subsystem attached to one [Hebb, 1949, Tulving, 1972, Gallistel and King, 2009].

The design consequence is sharp: the ledger may say what was surprising, and may therefore decide *what gets written and how deeply*, but it may not be descended. No gradient crosses a memory write. A node performs a write, not a fit.

2.2 Compression as the currency, without frequency as the organiser

We charge prequential codelength [Dawid, 1984, Rissanen, 1978, Blier and Ollivier, 2018], in the tradition that identifies compression with understanding [Solomonoff, 1964, Hutter, 2005]. But we deliberately do not organise memory by frequency, and the reason is a claim about scale rather than about statistics.

Remark 1 (Why counts degrade with data). A per-location count distribution $N(o | u)/N(u)$ is a frequency estimate. As a stream accumulates, the count distribution at any location converges toward the corpus marginal — it becomes the global unigram. A counted prior therefore *stops discriminating exactly as the data grows*: it degrades under scale, which is the same shape of failure as address crowding.

More data tends toward disorder; what makes it mean something is organisation, and organisation need not be a Bayesian prior [Jaynes, 2003]. The delta rule already *is* an organiser of the right kind: it is error-driven, so a row holds what needed correcting here rather than what was common here, and a token already predicted stops updating. Counts accumulate and degrade under growth; the delta rule converges and is stable under it. All counts and the escape chain were therefore removed, and the graceful degradation they used to provide was reassigned to the dynamics rather than to a smoothing rule.

2.3 An imprecise background is the point, not a compromise

The background is a fuzzy multi-timescale trace. It is meant to say *which memories are relevant here*, not to reconstruct what was said. We state as a non-goal that precise context association is not what this system is for: a design that makes the background exact has rebuilt the prefix

under another name, and inherits its costs. The nearest cognitive analogue is fuzzy-trace theory’s distinction between verbatim and gist representations [Brainerd and Reyna, 2002]; the nearest engineering analogue is that a multi-timescale reservoir is a context, not a transcript [Jaeger, 2001].

2.4 Time as a resource, and the ledger that must not be gameable

If depth is bought with the world’s silence, then a system’s efficiency is part of its capability, and evaluation must be conducted where silences are real. A closely related commitment concerns the shape of the world. If the agent’s actions *generated* its observations, surprise could be minimised by finding a quiet corner — the dark-room objection to naive free-energy accounts [Friston, 2010, Friston et al., 2012, Clark, 2013]. The escape is not a patch but a property of reality: the world proceeds whether or not the agent attends, so behaviour *influences* what is experienced without *generating* it, and missing something costs you later. We return to this in §9.

2.5 Awe, and the limits of a constructed test

We take seriously that intelligence is not a quantity our source can put under strain. A constructed stream can isolate mechanisms; it cannot decide whether those mechanisms amount to anything. We have tried to write the negative results of §7 as what they are — observations consistent with a plausible mechanism, on a source we argue cannot settle them — rather than as verdicts. Where the design should be judged is set out in §9, and we expect that judgement to be considerably harder than anything reported here.

3 Design

3.1 Notation and the algebra

Let d be the width, $\mathcal{C} = \{E_o\}_{o=1}^V \subset \mathbb{R}^d$ the fixed codebook, $\nu(\cdot)$ normalisation to the unit sphere, and $\otimes$ circular convolution,

$$(a \otimes b)_k = \sum_{j=0}^{d-1} a_j b_{(k-j) \bmod d}. \quad (1)$$

Write a^* for the involution $a_0^* = a_0$, $a_j^* = a_{d-j}$, and $m \oslash a = m \otimes a^*$ for circular correlation.

Definition 1 (Unitary codebook). $a \in \mathbb{R}^d$ is *unitary* if $|\hat{a}_j| = 1$ for every Fourier coefficient $\hat{a}_j$. We construct $\mathcal{C}$ by drawing phases uniformly and taking an inverse transform, so every E_o is unitary.

Proposition 1 (Exact inversion). *For unitary a , $a \otimes a^* = \delta$, hence $(a \otimes b) \oslash a = b$ exactly. For general a , $(a \otimes b) \oslash a = b \otimes (a \otimes a^*)$, and $a \otimes a^*$ is a δ only in expectation.*

Sketch. Convolution is multiplication in the Fourier domain and the involution conjugates, so $\widehat{a \otimes a^*}_j = |\hat{a}_j|^2$. This is identically one exactly when a is unitary; for Gaussian a the $|\hat{a}_j|^2$ are χ^2 -distributed with substantial variance, and the reconstruction is degraded before any interference. $\square$

Proposition 1 is not a technicality. Table 1 measures the cost: with a *single* triple stored and no interference whatever, a Gaussian codebook reconstructs at cosine 0.527.

3.2 The tick

Write x_{-1}, x_{-2} for the two previous observed tokens, c_i for the state of the i -th response, τ_i for the codebook identity it has resolved to, $w_i \in [0, 1]$ for how live it is, and $h(\cdot, \cdot)$ for a hash of two token identities into $\{1, \dots, B\}$.

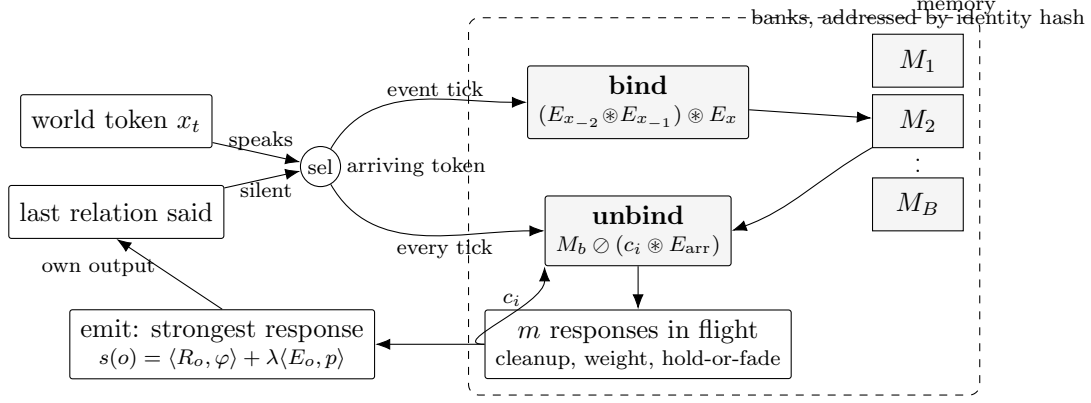


Figure 1: One tick. A token always arrives: the world’s when it speaks, the last relation it said when it does not. Speech binds a triple into the bank named by the two identities; every tick each response in flight unbinds one link out of the bank named by its own identity and the arriving one. The emission is the most matured response, scored by a learned row plus a direct codebook comparison, and it re-enters as the model’s own context. Nothing in the loop needs to know whether a question is outstanding.

Speech binds.

$$\mathcal{M}_{h(\tau_{x_{-2}}, \tau_{x_{-1}})} \leftarrow \nu((E_{x_{-2}} \otimes E_{x_{-1}}) \otimes E_{x_t}). \quad (2)$$

For a presented link $x \ r \ y$ the triple formed at the y tick is exactly $(E_x \otimes E_r) \otimes E_y$, and no episode boundary was needed to know that. The sum is *not* renormalised: $\mathcal{M} \leftarrow \nu(\mathcal{M} + t)$ shrinks everything already stored by $1/\sqrt{2}$ per write, giving the first item weight $2^{-k/2}$ — a sliding window, which is the one thing a superposition must not be.

Silence unbinds; and so does speech. On *every* tick each response advances one link:

$$q_i = \nu(c_i \otimes E_{\text{arr}}), \quad r_i = \nu(\mathcal{M}_{h(\tau_i, \tau_{\text{arr}})} \odot q_i), \quad (3)$$

$$\tau'_i = \arg \max_{o \in \mathcal{C}_{\text{seen}}} \langle E_o, r_i \rangle, \quad c_i \leftarrow E_{\tau'_i} \text{ iff } \langle E_{\tau'_i}, r_i \rangle > \theta(d, V). \quad (4)$$

Because $\otimes$ commutes, a response that has just reached a_1 after hearing r carries $E_{a_1} \otimes E_r$ as its next key with no bookkeeping. This is where the model’s own output does its work: the response *is* the position reached, and the relation stays on the table as the operator.

Arrival holds; speech supersedes.

$$w_i \leftarrow \begin{cases} 1 & \text{cleanup accepted,} \\ \gamma w_i & \text{rejected and the response has never advanced,} \\ w_i & \text{rejected and it has advanced at least once (it has arrived),} \end{cases} \quad (5)$$

and on every event tick $w_i \leftarrow \gamma w_i$ for all i before a new response is seeded at the observed token. From inside, “the chain ended” and “the chain never started” are identical; the only thing separating them is whether anything was ever retrieved. Holding a half-formed reply while the other person is quiet, and dropping it when they say something new, is the same picture.

The emission reads the most matured response. Let $i^* = \arg \max_i (w_i, \text{recency}_i)$ lexicographically. Then $p = c_{i^*}$, $\varphi = [p; \text{bound traces}; \text{event-locked bands}]$, and

$$s(o) = \underbrace{\langle R_o, \varphi \rangle}_{\text{learned}} + \lambda \underbrace{\langle E_o, p \rangle}_{\text{cleanup}}. \quad (6)$$

Remark 2 (Why not the mixture). Averaging unit responses and renormalising divides each by $\|\sum_i w_i c_i\| \approx \sqrt{\sum_i w_i^2}$. Measured, the response holding the answer arrived at the readout at 0.24–0.28 against a codebook floor of 0.255: present and indistinguishable from noise. “Several half-formed answers mature together and one of them is said” is not “say their average”. Reading the strongest is the natural readout of parallel responses, not a competition between mechanisms.

The write. Rows are updated by a delta rule $R_o += \eta (\mathcal{K}[o = x] - q(o)) \varphi$ over the target and a set of negatives. Those negatives are the tokens the model currently ranks highest, not a uniform sample: correcting what the model would actually have said is both error-driven and drawn from the same distribution the ledger charged. §6.2 shows this was the difference between a system that could represent a non-separable conjunction and one that could not.

3.3 Structural constraints

Five constraints are asserted by unit tests that run on every build.

- A1 Rate limit.** Constant computation per tick. The only way to think longer is to occupy more ticks, and ticks are given by the world.
- A2 Event-driven accounting.** Charged only when the world speaks, so “the identity input injects no new content” is a consequence of the ledger rather than an added axiom.
- A3 Read/write separation.** The write walk is unperturbed and content addressed; reads may wander.
- A4 No content write at baseline.** Identity ticks write nothing, asserted bit-for-bit over 500 baseline ticks.
- A5 Timescale reachability asymmetry.** The world writes all rungs; the model’s own output reaches only the fastest, so self-generated content contributes exactly zero to the slow timescales — a structural ceiling on confabulation.

3.4 The background, and why its key is frozen

The background is a cascaded band-pass ladder: one use of a ladder, on the time axis only. The memory itself is flat, because a ladder inside the store would be an abuse of the device.

The bands are snapshotted *at the last event*, so the retrieval key stops moving when the world does. The reason is that the bands carry two things at once: which regime this is, which must hold across a silence, and how long the world has been quiet, which changes every tick. With both in the key, elapsed time decides whether a memory fires at all — measured, a fact learned at six ticks of silence did not answer at one, scoring 0.000 where the same pair presented as an ordinary fact scored 0.74. A clock in the key is a bug; the live fast rung remains available to the commit rule, which is where elapsed time belongs.

4 Isolated verification

The evidence in this section is from unit tests on synthetic memories in which every quantity is known. It says what the mechanism does. It says nothing about whether the mechanism is useful on any data. We keep the two kinds of evidence in separate sections throughout, because conflating them is the easiest way to overstate a result of this kind.

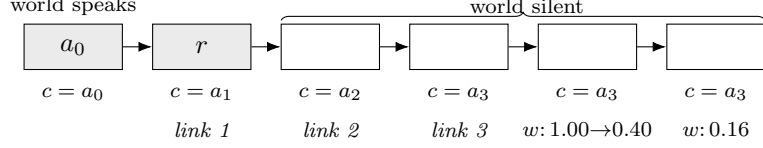


Figure 2: A walk $a_0 \rightarrow a_1 \rightarrow a_2 \rightarrow a_3$ presented forty times among unrelated traffic, then asked with the start and the relation. One link per tick; the end of the chain has no outgoing link, so the unbinding returns noise, cleanup refuses it, and the response holds its answer rather than stepping past it. Surplus thinking time is harmless. The relation’s own response fades on the same rule.

4.1 One link per quiet tick

Four things had to be right simultaneously for Figure 2, and each was found by a four-second test rather than by an end-to-end run (§8): the relation is the operator and the model’s own output is the position, not the reverse; seeding must retire the *least live* response rather than the oldest, because on ties “oldest” is the slot holding the response that has just advanced; responses must be weighted by whether they are still advancing; and the configuration must lie inside the operating region below.

4.2 Capacity, and the operating region

Proposition 2 (Operating region). *Let $\mathcal{M} = \sum_{i \leq k} t_i$ with t_i unit and mutually near-orthogonal, and let cleanup take the best match among V codebook entries in $\mathbb{R}^d$. Then a stored triple returns $\langle \hat{\mathcal{M}}, t_j \rangle \approx k^{-1/2}$, an unwritten key returns $\approx d^{-1/2}$, and a random vector’s best codebook match is $\approx \sqrt{2 \ln V / d}$. Retrieval is usable when*

$$\frac{1}{\sqrt{k}} > \sqrt{\frac{2 \ln V}{d}} \iff d > 2k \ln V. \quad (7)$$

At $d = 256$ we measure cosine 0.043 with $k = 401$ against a predicted 0.050. Equation (7) is what gives content addressing a job that is not about accuracy: the address chooses *which* superposition to unbind, each bank stays under capacity, and the read stays one unbinding however much has been stored. Related capacity analyses for distributed representations appear in Plate [1995] and Frady et al. [2018].

It also fixes the cleanup threshold, which must not be a hand-set constant: $\theta(d, V) = \kappa \sqrt{2 \ln V / d}$. Ours was 0.25 against a floor of 0.452 — *below* the noise — so cleanup accepted noise on 97.6% of silent ticks and the response was randomised every tick while the instrument reported near-perfect success.

codebook	$k = 1$	$k = 16$	$k = 64$
Gaussian, unit norm	0.527	0.205	0.120
unitary (random phase)	1.000	0.251	0.127
$k^{-1/2}$ (theory)	1.000	0.250	0.125

Table 1: Reconstruction cosine, $d = 256$, mean of 12 draws. Proposition 1 in numbers: a Gaussian codebook loses almost half the signal with a *single* item stored, and Proposition 2 is a statement about unitary vectors. This had been assumed rather than checked for the life of the project.

4.3 Addressing must not preserve locality

A key nobody wrote still hashes into an occupied bank, and its occupants answer. The end of a chain therefore does not retrieve *nothing*; it retrieves a neighbour. No threshold separates the two, because both are real stored content of comparable strength. This is address crowding [Yang, 2026] in its sharpest form.

We built a read-back check for it — bind the candidate back onto the key and ask whether that triple is in the bank at all — and it failed instructively.

Lemma 1 (Locality defeats verification). *Under sign-bit locality-sensitive hashing [Indyk and Motwani, 1998, Charikar, 2002], the occupants of a bank have keys close to the query. If cleanup returns some occupant’s target y_i then, for unitary vectors,*

$$\langle \nu(p_i \otimes y_i), \nu(q \otimes y_i) \rangle = \langle p_i, q \rangle,$$

and $\langle p_i, q \rangle$ being large is precisely why p_i and q share a bank. The verification is defeated by the property it was built to detect.

Addressing needs same-content-same-bank and gains nothing from similar-content-similar-bank; the second property is pure cost and it is what makes a near miss undetectable. Scrambling the code removes the locality but makes the address brittle, since write and read reach the same convolution by different floating-point paths and one flipped sign then means an unrelated bank. That dilemma is false: cleanup has already resolved the response to an exact codebook token and the write holds both identities, so the address can be a hash of the *identities*. Vectors carry the superposition; identities carry the address. With that in place the read-back check becomes inert — zero rejections — and we report it as doing nothing rather than crediting it.

5 The source and its instruments

Regimes arrive, run, and depart permanently; no task identifier, no boundary signal, no reset. Each carries first-order facts, a Latin square, a product code, composition chains, and walks. Two families can only be answered by computing: composition queries are posed *once* per chain and withheld Latin cells are asked once each, so nothing about them can be memorised from a previous presentation.

The generator is self-tested before any model is built, because on this project it produced as many serious defects as the model did.

quantity	Latin square	product code
$I(T; A \mid D), I(T; B \mid D)$ (bits)	0.126 / 0.127	non-zero by design
prequential single-cue baseline	0.201 / 0.195	0.486 / 0.402
analytic value	$1/5 = 0.200$	0.500
composition leakage		0
facts $\times$ / practice per fact $\times$ at $3\times$ load	2.6–3.0 / 1.00–1.14	

Table 2: Source self-check, measured on the emitted stream with no threshold. Marginal information uses the Miller–Madow correction [Miller, 1955]. The single-cue baseline is *prequential* — predict from counts so far, then update — because an in-sample ceiling is an oracle that no model charged before it sees the answer can reach; the in-sample figure is 0.27 where the prequential one is 0.20.

An earlier version drew one Latin cue from a Zipf distribution and the other uniformly. For $T = (a+b) \bmod m$, $P(T \mid A)$ is uniform exactly when p_B is, so that square was *not* zero-marginal: a predictor using cue B alone reached 0.2745, and every Latin accuracy the project had reported

was at or below it. Both cues are now uniform, and it must be both — no choice of square repairs it, and making both Zipf only makes the leak symmetric. The frequency skew now lives outside the square, where it forges no shortcut.

6 End-to-end measurements

The evidence in this section is from the full system on the synthetic stream: 90k ticks, 12 regimes, one seed unless stated. It is source-dependent.

6.1 Retrieval occurs, and holds

arm	t_0	t_1	t_2	t_3	t_4	t_5
control (no superposition)	0.014	0.015	0.015	0.019	0.021	0.021
4096 banks, 4 responses	0.014	0.185	0.240	0.268	0.237	0.238
8192 banks, 4 responses	0.014	0.189	0.243	0.275	0.248	0.250
4096 banks, 1 response	0.006	0.193	0.007	0.010	0.013	0.013

Table 3: Cosine between the emitted response state and the correct answer, by tick of silence; t_0 is the relation’s own tick. Retrieval happens and holds. With a single trajectory it happens and then collapses — the cleanest evidence we have for the “not just one trajectory” clause.

6.2 Competitive negatives

The largest single effect we measured was not a design choice but the repair of a defect. The delta rule drew negatives uniformly from every known token. Discrimination on a Latin item rests entirely on pushing down the five rivals sharing a square’s six targets; each was drawn about 2.3% of the time against a target appearing roughly eighteen times in a regime’s life, so a rival received a negative gradient about 0.4 times. The readout was never taught to separate the six and settled at 1/6 — and, because the product code is answerable from marginals where positive-only ranking suffices, the separable/non-separable split that had organised months of analysis was a property of the sampler rather than of the architecture.

negatives	Latin	product	one-shot composition	answer bits
uniform, $k=16$	0.1448	0.6884	0.0150	6.249
uniform, $k=64$	0.2421	0.7612	0.0150	5.173
top- k , $k=16$	0.4529	0.8426	0.3308	3.970
top- k , $k=64$	0.4444	0.8426	0.3459	3.984

Table 4: Correcting what the model would actually have said rather than random tokens [Mikolov et al., 2013]. Raising the uniform count moves the same way and falls well short, which is the control saying this is about *which* negatives and not how many. Composition goes from zero to a third: chaining two stored facts was not absent, it was unreachable.

6.3 Exposure, and what it says about sources of this shape

Table 5 also exposes a representational fact. A novel product pair scores 0.000 where cue A alone would give 0.5: $E_a \otimes E_b$ is a lookup key, not a decomposable representation, so an unseen combination is a fresh random vector uncorrelated with anything stored. We have the *combine*

family	1st exposure	2nd	3rd	4th+
Latin ($n = 286$ firsts)	0.000	0.03	0.33	0.87
product ($n = 72$)	0.000	0.02	0.37	0.995
composition ($n = 35$)	0.000	0.00	0.23	0.95

Table 5: Accuracy by how many times a fact had already been charged, under an earlier configuration. Every family is exactly zero the first time and near ceiling by the fourth: every aggregate on a present-repeat-ask source is carried by re-lookup. This is why composition queries and withheld cells are now asked *once*.

half of distinguish-and-combine, and a learned linear readout cannot recover the parts. That is exactly the gap the unbinding half of §3 exists to close, and closing it is what moved composition from 0.015 to 0.331 in Table 4.

6.4 Aggregates

arm	Latin	product	retention	answer bits	walk support
no superposition	0.5695	0.8273	0.4187	3.141	0.517
baseline	0.6090	0.8290	0.4405	3.060	0.546
banks = 1	0.5679	0.8282	0.4464	3.051	0.540
banks = 65536	0.6107	0.8290	0.4306	3.056	0.546
bind off (<i>must collapse</i>)	0.1809	0.7029	0.2996	4.577	0.219

Table 6: Single seed; codelength charged at the answer tick only. Against the single-cue baselines of Table 2 (0.20 and 0.49) both conjunction families are answered well above a predictor that ignores half the challenge. The collapse arm holds, which is what makes the rest readable.

7 What this source cannot settle

Three components measure net-negative here: the ladder feedback of the model’s own channels, the operator graph, and background depth beyond one rung. One mechanism explains all three and the arms order monotonically by it. We rule on none of them.

arm	Latin	retention	answer bits	walk $d=1$	non-fact content in φ
all streams off	0.6147	0.4802	2.943	[.286 .222 .667 .200]	least
baseline	0.6090	0.4405	3.060	[.000 .111 .333 .000]	
live bands	0.6123	0.4008	3.138	[.000 .000 .333 .000]	+ a clock
graph re-enabled	0.5913	0.3413	3.428	[.000 .000 .333 .000]	most
rungs = 1	0.5751	0.6111	3.359	[.286 .000 .333 .200]	fewest bands
rungs = 5	0.5775	0.4702	2.948	[.000 .222 .333 .200]	most bands

Table 7: Single seed. Retention age-decay slopes: -0.023 at one rung against -0.072 at the baseline’s three. Walk cells hold six to nine events and are not a strong result.

The hypothesis. φ is a fixed-width retrieval key. Anything added to it that is not a function of the fact being asked about adds variance without signal. On this source a Latin answer is determined by its two cues, so what the model just said carries no information about it while

making φ differ between two presentations of the same fact. The operator graph is near-random for the same reason its learning does nothing: each edge receives ten to forty rank-one updates against 4096 parameters, and its drift grows as $\sqrt{\text{updates}}$ — the signature of uncorrelated accumulation. Bands are context, not fact.

Why we do not rule. Every one of these rests on the source’s answers being cue-determined, which is the property of a constructed source most open to doubt. The hypothesis predicts a sign flip on a source where the answer is *not* cue-determined and the model’s own trajectory carries necessary information; the only family here with that property has cells of six to nine events.

The three-stream requirement has two implementations with opposite signs. Ladder feedback carries no signal here and is net-negative. Cursor advance — the model’s own output as the position a chain has reached — is load-bearing (Table 3). The founding requirement that input and output streams interleave as context is met by the second; the first is one way of writing it that happens to hurt here. Reporting them as a single verdict would be wrong.

A test we abandoned. We proposed measuring the variance of φ across repeated presentations of the same fact. It has almost no power: 2304 dimensions, four to nine presentations per fact, and a rank correlation between two noisy orderings. Two difference tests replace it — excluding a component from φ while leaving it in the dynamics, and substituting a noise block of matched width and variability — but both belong on real data (§9).

8 Method: five failures of one shape

symptom	cause	what found it
self-stream ablation reads zero	feature blocks wider than rows; three background bands silently truncated	per-block alignment assertion
near-miss curve flat	the write path was dead code; reads and writes landed on the same perturbed edge	call-graph grep
every accuracy collapses	a decay constant applied <i>per tick</i> , leaving 1.6% of the binding by the time it was read	the mechanism’s own off-control
six arms bit-identical	the retrieval never reached the emission	the bit-identity itself
cleanup succeeds 97.6% and is always wrong	the threshold sat <i>below</i> the codebook noise floor (0.25 against 0.452)	measuring four numbers directly

Table 8: Each symptom reads as “a mechanism measures inert”. Each cause is that the mechanism’s product had no outlet, or was erased downstream.

The discipline this forces, stated because we expect it to generalise to any system of this shape:

1. Any change on the path of a load-bearing mechanism gets that mechanism’s off-control *in the same sitting*. If the full arm and the off arm are close, stop.
2. A terminal metric cannot debug a multi-step chain. An end-to-end zero has no power to say which of five steps broke: four-second unit tests found four errors where ten-minute end-to-end runs found none.

3. Bit-identical arms are more alarming than bad numbers.
4. Any hand-set constant must be back-solved from its own units. Three of our failures were constants never checked against the scale they operate on.
5. Suspect the data before the mechanism.

Performance belongs in the same list, because where computation is bought with the world’s time it is not merely engineering. Replacing an address that regenerated 4096 random keys per call with pre-generated sign bits took one arm from over twenty-one minutes to 6m43s, and the $O(d^2)$ convolution became $O(d \log d)$, which is also what makes the unitary codebook affordable.

9 Limitations, and what a real evaluation requires

A replayed stream cannot test a model for which time is a resource. Our silences are drawn from $\{1, 2, 4, 8\}$ — authored, not observed. The model is never late: it receives exactly the ticks the file specifies regardless of how much computation it needed. There is no rate, and in a real continual setting the ratio between the world’s event rate and the agent’s compute rate is decisive. Replay tests mechanism; it cannot test the claim this architecture exists for.

The autoregressive baseline is the wrong yardstick and the right lower bound. We report PPM [Cleary and Witten, 1984] not as a standard to beat — that adopts the next-symbol objective as the measure — but as a control that *cannot accumulate*: a bounded window carries nothing across itself, so a sustained gap is evidence of accumulation. On charged events, order-4 PPM costs 11.80 bits at accuracy 0.008 and scores exactly 0.000 on both conjunction families, against 9.17 bits and 0.206/0.732 for the mechanism at the time of that comparison. The variant handed the segmentation the mechanism refuses to assume reaches 5.60 bits, and losing to that is the comparison worth losing.

Our source has no transferable structure. Each regime carries its own square, entities and product code and shares nothing with any other, so transfer is impossible by construction and an acquisition-acceleration measurement — is the n -th regime learned faster than the first? — is guaranteed flat. If a constructed source is retained, the one change worth making first is shared latent relations across regimes, because that is the minimum condition under which organisation has anything to organise.

Attention is empty without laws. Letting the model’s state influence which of the concurrently live regimes speaks next — $P \propto \exp(\beta \cdot \text{align})$, with $\beta = 0$ recovering the present source exactly — is meaningful only when the same regularity recurs across situations, so that attending here teaches you about there. The world’s progression must remain exogenous: things happen whether or not the agent attends, and that property alone stops the ledger being gamed by an agent that finds a quiet corner [Friston et al., 2012]. Real reactions are influenced by the individual without being generated by it, and requiring a full self-loop is the wrong route.

What would settle these questions. Real language content carrying real, non-authored temporal structure: timestamped transcripts, subtitle streams, conversation logs — data in which silences are real silences of real duration and the two-people-talking picture is the data rather than an analogy. A tick then corresponds to a unit of real time, and the natural next step binds the compute budget to wall-clock, at which point depth-from-time becomes a genuine constraint and the efficiency of an implementation becomes part of its capability. This is where LeCun [2022] and the embodied tradition [Brooks, 1991, Clark, 2013] both point.

Caution. We do not know what these mechanisms do on real data and do not think it can be predicted from here. Every result is single-seed unless stated, and the families that cannot be memorised carry sixty to ninety-six events in total.

10 Related work

Distributed binding. Our store is a superposition read by correlation, in the holographic [Plate, 1995] and tensor-product [Smolensky, 1990] traditions, addressed the way sparse distributed memory addresses [Kanerva, 1988], and surveyed as hyperdimensional computing [Kanerva, 2009, Gayler, 2003, Eliasmith, 2013, Kleyko et al., 2023]. Capacity of such stores is analysed by Plate [1995] and Frady et al. [2018]; our Proposition 2 is the form we needed, coupling capacity to a cleanup codebook of size V . Classical associative memories [Willshaw et al., 1969, Hopfield, 1982, Kohonen, 1972] are the ancestors of the readout.

Learned versus fixed addressing. Differentiable memories [Graves et al., 2014, 2016, Weston et al., 2015] and modern associative layers [Ramsauer et al., 2021] learn where things go. Ours is fixed by construction, following Yang [2026]: an address that no later learning can move is what buys freedom from interference, and the price is retrieval difficulty. Fast-weight schemes [Schmidhuber, 1992, Ba et al., 2016] also write associations at inference time, but inside an autoregressive objective.

Random transforms. The graph component turned out to behave as a random feature expansion [Rahimi and Recht, 2007] feeding a linear readout, which is the regime reservoir computing occupies deliberately [Jaeger, 2001, Maass et al., 2002]. We report this as a negative structural finding rather than as a design.

Continual learning. The setting and its interference are classical [McCloskey and Cohen, 1989, French, 1999]; responses include regularisation [Kirkpatrick et al., 2017], replay [Lopez-Paz and Ranzato, 2017], and architectural allocation, surveyed by Parisi et al. [2019]. Our position is the allocation one, taken to its committed form.

Compression and prediction. The ledger is prequential [Dawid, 1984, Rissanen, 1978] in the description-length reading of learning [Blier and Ollivier, 2018, Grünwald, 2007], and the identification of compression with understanding is argued at length by Solomonoff [1964] and Hutter [2005]. What we do not take is the objective: there is no reward here [Sutton and Barto, 2018, Silver et al., 2021].

Autoregressive systems can do these tasks. A long-context model can solve retrieval of this shape by induction [Olsson et al., 2022], and state-space models [Gu and Dao, 2023] carry a running state rather than a window. Neither can spend a lull: computation per token is fixed, and that is the axis on which this work differs.

11 Conclusion

We built a challenge-response memory in which binding and unbinding are the two halves of one tick, selected by whether the world spoke, and in which several partial responses mature in parallel while the world is quiet. In isolation the mechanism does what the design says: one link per quiet tick, a halt at the end of a chain, surplus silence harmless. We derived the operating region of its store, showed that binding inverts exactly only for unitary codebooks, and showed

that locality-sensitive addressing makes a near miss undetectable by construction — so half of the crowding penalty this family is known for was, in our implementation, self-inflicted.

On a synthetic continual stream retrieval occurs, holds, and requires several trajectories rather than one. Several components measure net-negative with one mechanism that explains them; we rule on none, because the explanation rests on a property of the source rather than of the world, and because a replayed file cannot test a system for which time is a resource. What we claim is feasibility at this scale on this source, for a design that is non-autoregressive at its root, that treats precise context association as a non-goal, and that is aimed at stateful continual learning in a world that has a clock. Where it should be judged is on real language carrying real time, and we expect that judgement to be substantially harder than anything reported here.

Reproducibility

The implementation is 8953 lines of Rust with no dependencies, deterministic under a counter-based generator with fixed-order float reductions, carrying 20 structural assertions and 6 generator self-tests that run on every build. Two identical runs agree bit-for-bit. Constraints A1–A5 and each mechanism claim of §4 are asserted by a named test.

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